

PAPER

A Reconciliation Design for Secret-Key Throughput Enhancement in Real-time Satellite FSO/QKD

Cuong T. NGUYEN[†], *Nonmember*, Hoang D. LE[†], and Anh T. PHAM^{†a)}, *Members*

SUMMARY Satellite-based free-space optics (FSO)/quantum key distribution (QKD) systems have aroused considerable attention due to their ability to enable quantum-safe global/wireless applications. For real-time QKD applications, a critical challenge is achieving a high secret-key throughput (SKT), defined as the average number of secret bits produced per second. Among the stages in the QKD process, key reconciliation (KR) is widely recognized as a major throughput bottleneck. In this paper, we introduce a novel KR strategy based on blind reconciliation and the rate-compatible (RC) code family to enhance the SKT of satellite-based FSO/QKD systems. The key idea is to determine an optimal initial code rate that balances KR efficiency and the number of communication rounds. For this purpose, we develop a comprehensive analytical framework in which the cumulative distribution function (CDF) of the quantum bit-error rate (QBER) across blocks is first derived. Based on this derivation, a closed-form expression for the SKT is obtained, enabling the optimal initial code rate selection. Simulation results confirm the effectiveness of our proposed design over conventional approaches under various fading conditions.

key words: Quantum key distribution (QKD), free-space optics (FSO), key reconciliation (KR), error estimation, quantum bit-error rate (QBER)

1. Introduction

Most of today's communication systems rely on public-key cryptography (PKC) to share secret keys serving confidential data transmissions. The security of PKC relies on the hardness of solving specific mathematical problems within the useful lifetime of the information. However, recent years have witnessed advances in computational technologies, notably in quantum computers capable of executing quantum algorithms (e.g., Shor's algorithm). This poses a growing threat to PKC-based key distribution, necessitating research into quantum-safe solutions. A viable candidate is quantum key distribution (QKD), which is based on quantum mechanics to provide information-theoretic security [1]. Experimental validation of QKD protocols has been successfully conducted across various scenarios, such as fiber or free-space optical (FSO) systems. To enable quantum-safe global/wireless applications, e.g., secure Internet of Vehicles, satellite-based FSO/QKD systems have emerged as a potential solution.

A QKD procedure consists of two phases: the quantum phase and the post-processing phase, operating over the quantum channel and the classical public channel, respectively. In real-time satellite QKD systems, it is critical

that neither phase becomes a performance bottleneck, as this would reduce the secret-key throughput (SKT), defined as the average number of secret bits generated per second. In recent years, extensive work has focused on increasing the distribution rate of key materials over the quantum phase. For example, quantum sources of discrete-variable (DV)-QKD featuring gigahertz pulse rates were demonstrated for satellite downlink scenarios [2]. Moreover, the potential of continuous-variable (CV)-QKD in improving the secret key rate has also been discussed and investigated [3–5]. Modulation-signaling QKD protocols, which can achieve Mbps-level key-generation rates, have also been considered for space-based QKD settings [6]. The increasing speed of the quantum phase requires a corresponding enhancement in the processing capability of the post-processing phase.

Meanwhile, key reconciliation (KR), a step after sifting in the post-processing phase, is often regarded as a throughput bottleneck in QKD systems [7]. The role of KR is to eliminate discrepancies introduced by imperfect hardware and/or eavesdropping attempts between the two users' sifted bits. These disturbances are characterized by the quantum bit-error rate (QBER). A common KR approach relies on the well-known low-density parity-check (LDPC) codes [8]. Specifically, Alice[†] treats a block of sifted bits as a codeword and generates side information using an LDPC code, which are normally syndrome bits. Upon receiving these bits via the public channel, Bob leverages the syndrome-based decoding algorithm to correct his sifted block accordingly [9]. The importance of the KR step can also be recognized via recent extensive research efforts on fiber and terrestrial FSO/QKD systems, e.g., [10–13]. This motivates us to focus on KR designs of real-time FSO/QKD systems.

Existing works on KR design in satellite-based FSO/QKD systems have primarily focused on the rate-compatible (RC)-LDPC code family to cope with fluctuating QBER due to the FSO channels [14–17]. These structures provide nested code members, implying that the syndrome bits of a high-rate code are part of a lower-rate code. This allows code-rate transitions without leaking extra information, making it a viable solution when QBER across sifted blocks fluctuates widely. Notably, the designs of rate-adaptive KR with RC-LDPC codes constructed via shortening and puncturing (*sp*) were reported in [14–16]. Using the rate-adaptive KR approach, both sides first estimate the error through *ran-*

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[†]Department of Computer Science and Engineering, The University of Aizu, Fukushima 965-8580, Japan

a) E-mail: pham@u-aizu.ac.jp

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[†]For the rest of the paper, we assume that Alice is the one who sends the syndrome bits without loss of generality.

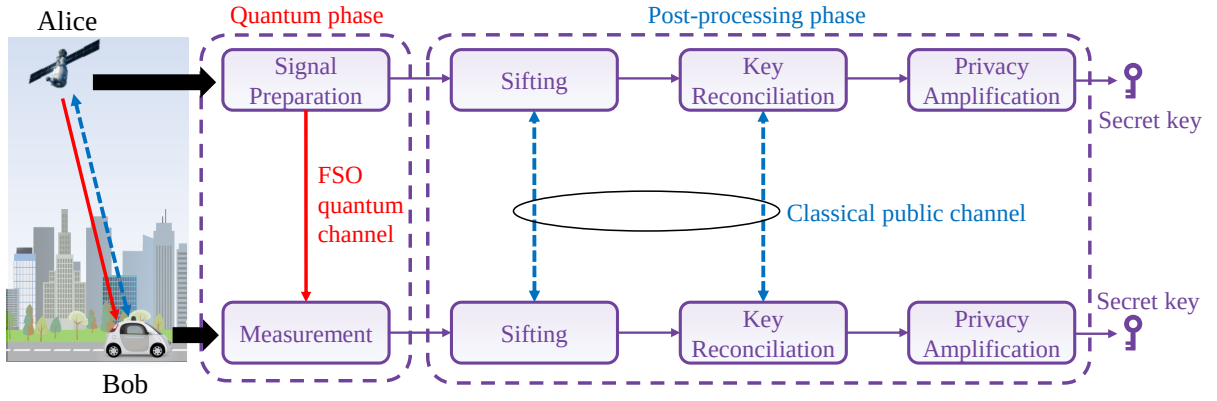


Fig. 1 Block diagram of considered FSO/QKD systems, including the quantum phase and the post-processing phase.

dom sampling, by revealing and comparing a subset of their sifted blocks via the public channel. Based on this estimate, a proper code rate is determined from the RC-LDPC code family. Alternatively, *blind KR* eliminates the need for a prior error estimation step. Our recent work proposed a blind KR scheme based on protograph-based RC-LDPC codes [17]. This approach initializes with the highest-rate code and progressively reduces the code rate after each failed decoding attempt until successful decoding is achieved or all available rates are exhausted.

It is worth noting that the blind KR method has been widely discussed in the literature [18–22]. In [18], the authors consider a blind KR design with *sp*-based RC-LDPC code. In particular, Alice adds random auxiliary bits to her sifted block to form an extended block. Alice then transmits the syndrome bits of the extended block to Bob. Since these auxiliary bits are unknown to Bob, he treats them as punctured bits during decoding. By randomly disclosing the values of auxiliary bits from the extended block, so-called shortened bits, Alice can arbitrarily adjust the code rate. The study [19] proposed an enhanced blind KR protocol that improves reconciliation efficiency and reduces the mean number of communication rounds. This is achieved through a symmetric operational framework and an efficient method for disclosing auxiliary bits based on failed decoding results.

In [20], the authors proposed a blind KR scheme with incremental step sizes to accelerate the processing time. The key idea is that Alice progressively increases the number of revealed shortened bits each round, instead of employing a fixed number as in [18]. The study [21] pointed out a drawback of *sp*-based RC-LDPC codes, which is the limited range of effective code rates. This results in a deterioration in reconciliation efficiency when QBER values are highly fluctuating. To tackle this problem, the authors proposed a polar-code-based blind KR design. This limitation was later addressed for LDPC codes in [22]. Specifically, this work utilized protograph LDPC codes to construct RC code families with a better reconciliation efficiency than *sp*-based ones over a wide range of QBER. The protograph LDPC codes are widely known for their low-complexity structure

and near-capacity performance [23]. This type of LDPC code was also considered in [17].

Nevertheless, existing methods may fall short in providing a satisfactory SKT in high-latency satellite-based FSO/QKD systems. In particular, rate-adaptive KR relies on accurate EE. A popular EE approach is random sampling, which requires revealing a portion of sifted bits that must be discarded afterward, thereby reducing SKT. Moreover, this approach incurs an additional communication round for information exchange, further degrading SKT, especially when the average QBER is low. Another EE approach uses the QBER values from previous blocks to predict the error rate of the current block [24]. This method may be ineffective in the desired systems, where the QBER varies significantly across blocks. Syndrome-based EE, which exploits the syndrome vector to estimate errors, can achieve acceptable accuracy while avoiding the disclosure of sifted bits [25]. Such approaches typically initialize decoding with the highest-rate code, which may not be optimal when the average QBER is high, as the initial decoding attempts are likely to fail. The limitations of EE, particularly in satellite-based FSO/QKD systems, motivate the adoption of blind KR. This, in turn, necessitates an efficient design of blind KR tailored to such systems. It is noted that existing blind KR schemes may suffer from low SKT under highly fluctuating QBER and long-distance satellite links, as multiple communication rounds are typically required. Given the large round-trip times in these systems, such iterative decoding significantly increases post-processing latency, thereby degrading the achievable SKT.

This paper proposes a blind reconciliation-based KR strategy to enhance the SKT of satellite-based FSO/QKD systems. Our proposal adopts the blind KR mechanism with protograph RC-LDPC codes to cover a wide range of QBER, as in [17, 22]. *However, unlike these works, our key idea is to optimally determine the initial code rate based on the statistical distribution of the QBER, thus improving the SKT of the considered system.* To this end, we derive an analytical approximation of the cumulative distribution function (CDF) of the QBER across sifted blocks. Based on this derivation, a

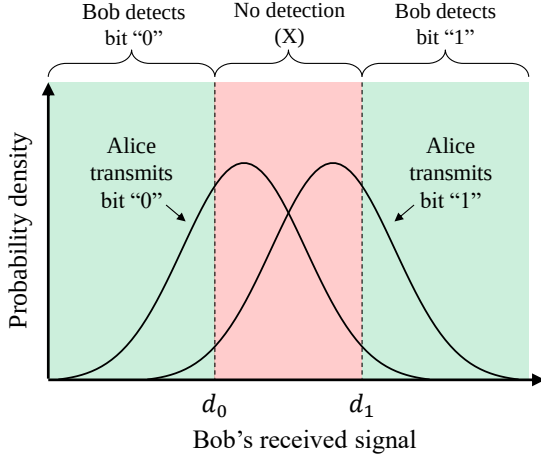


Fig. 2 PDF of Bob's received signal with three possible detection outcomes.

closed-form expression of SKT is obtained, enabling optimal selection of the initial code rate and reducing unnecessary decoding rounds. Numerical results demonstrate that the proposed scheme significantly improves SKT compared with state-of-the-art approaches. Furthermore, its performance is validated under various channel conditions and system parameters.

The rest of the paper is structured as follows. Sec. 2 describes the system overview, the channel, and the signal model. Sec. 3 elaborates on the proposed blind KR strategy. Numerical results and discussions are presented in Sec. 4, with concluding remarks provided in Sec. 5.

2. System Descriptions

2.1 System Overview

Fig. 1 illustrates the considered FSO/QKD system, where a satellite (Alice) attempts to share secret keys with a ground vehicle (Bob). We consider the free-space CV-QKD protocol using subcarrier (SC) binary phase-shift keying (BPSK) and dual-threshold/direct detection (DT/DD) [26]. This protocol adapts the concept of the well-known BB84 protocol, yet utilizes conventional optical transceivers, to provide a reduced-complexity solution. In addition, we consider the unauthorized receiver attack (URA) model involving a mobile eavesdropper (Eve), such as a ground or aerial vehicle. Particularly, Eve tries to tap the transmitted FSO signals by being within the beam footprint while simultaneously monitoring the classical public channel.

The key distribution process can be divided into two phases, i.e., the quantum phase and the post-processing phase. In the *quantum phase*, Alice first transmits random binary bits modulated by the SC-BPSK with a small modulation depth δ over the atmospheric FSO links. At the receiver, Bob directly detects the signals with dual thresholds, i.e., d_0 and d_1 . Fig. 2 illustrates the PDF of receiver signals, affected by additive white Gaussian (AWGN) noise, in cases where

bits "0" and "1" are transmitted. As shown, Bob detects bits "0" and "1" when the received signals are below d_0 and above d_1 , respectively. Otherwise, he detects no bits, i.e., "X". After gathering ℓ_b bits, Bob starts the *sifting* step by announcing the timestamps at which he detects these bits. Alice discards her bits accordingly. Eventually, both sides form their own *sifted blocks*.

Noting that the sifted blocks are partially correlated. As a result, the blocks are passed to the *key reconciliation* (KR) stage to remove mismatches. The KR stage sequentially processes the sifted bits in discrete blocks of size ℓ_b . In this work, we consider blind reconciliation, which is detailed in Sect. 3. Sifted blocks that fail the reconciliation are discarded, while successfully reconciled blocks are stored for the subsequent *privacy amplification*. Specifically, after reaching a certain number of blocks, both users independently concatenate them into a longer privacy amplification block. Then, they apply a preselected hash function from a universal hash family to produce a shorter, yet more secure, key [27].

2.2 Quantum Channel and Signal Models

The channel model characterizes the joint effects of three impairments, i.e., cloud attenuation h_c , atmospheric turbulence h_t , and pointing error h_p . As for the cloud attenuation, it is expressed as [28]

$$h_c = \exp[-\sigma H_c \sec(\xi)], \quad (1)$$

where H_c is the cloud vertical height, ξ denotes the zenith angle, and σ is the attenuation coefficient, which can be computed via the optical wavelength λ and the visibility V as in [28, (12)]. The visibility is given as $V = 1.002/(M_c N_c)^{0.6473}$, where M_c (g/m^{-3}) denotes the cloud liquid water content (CLWC), and N_c (cm^{-3}) represents the cloud droplet number concentration. Regarding the *atmospheric turbulence*, its probability density function (PDF) can be described by a Gamma-Gamma (GG) distribution for moderate-to-strong turbulence conditions as [6]

$$f_{\text{GG}}(h_t) = \frac{2(\alpha\beta)^{\frac{\alpha+\beta}{2}}}{\Gamma(\alpha)\Gamma(\beta)} h_t^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}(2\sqrt{\alpha\beta}h_t), \quad (2)$$

where $\Gamma(\cdot)$ denotes the Gamma function, and $K_{\alpha-\beta}(\cdot)$ is the modified Bessel function of the second kind of order $(\alpha - \beta)$. Furthermore, α and β are effective numbers of large- and small-scale eddies, which are calculated via the Rytov variance σ_R^2 , as in [6]. The computation of σ_R^2 can be found in [28, (15)].

Regarding the *pointing error coefficient*, its PDF in the general case can be mathematically represented as [6]

$$f_{h_p}(h_p) = \frac{\varphi_{\text{mod}}^2}{A_{\text{mod}}^2} h_p^{\varphi_{\text{mod}}^2-1}, \quad 0 \leq h_p \leq A_{\text{mod}}, \quad (3)$$

where $\varphi_{\text{mod}} = w_{\text{Leq}}/2\sigma_{\text{mod}}$, w_{Leq} is the equivalent beam

width and computed as a function of the effective beam width $w_{L,\text{eff}}$ and the receiver aperture radius r_a , as in [28]. In addition, $w_{L,\text{eff}} \approx w_0 \sqrt{1 + \left[2L / (k_w w_0^2)\right]^2}$, where $L = (H_s - H_v) \sec(\xi)$ is the slant distance, H_s is the satellite altitude, H_v is the vehicle altitude, $k_w = 2\pi/\lambda$, $w_0 = \lambda/(\pi\theta)$, and θ is the divergence half-angle [29]. Moreover, $\sigma_{\text{mod}} = \left[(3\mu_x^2\sigma_x^4 + 3\mu_y^2\sigma_y^4 + \sigma_x^6 + \sigma_y^6) / 2 \right]^{1/3}$ is the modified variance, where $\mu_{x/y}$ and $\sigma_{x/y}$ are the means and standard deviations of jitter along x/y -axis, respectively. Finally, $A_{\text{mod}} = A_0 \exp \left(\frac{1}{\varphi_{\text{mod}}^2} - \frac{1}{2\varphi_x^2} - \frac{1}{2\varphi_y^2} - \frac{\mu_x^2}{2\sigma_x^2\varphi_x^2} - \frac{\mu_y^2}{2\sigma_y^2\varphi_y^2} \right)$, where $\varphi_\varsigma = w_{L,\text{eq}}/2\sigma_\varsigma$, $\varsigma \in \{\text{mod}, x, y\}$, and A_0 can be computed as in [28]. Regarding the composite channel model, the numerically tractable expression of its PDF can be written as [6, (14)]

$$f_h(h) = \frac{\tilde{\varphi}}{\chi \tilde{\varphi}} h^{\tilde{\varphi}-1} \sum_{i=1}^n a_i \psi_i \tilde{\varphi}^{-\alpha} \Gamma \left(\alpha - \tilde{\varphi}, \frac{\psi_i h}{\chi} \right), \quad (4)$$

where $\tilde{\varphi} = \varphi_{\text{mod}}^2$, $\chi = A_{\text{mod}} h_c$, $\Gamma(\cdot, \cdot)$ is the upper incomplete Gamma function, $a_i = \frac{\theta_i}{\sum_{j=1}^n \theta_j \Gamma(\alpha) \psi_j^{-\alpha}}$, $\psi_i = \frac{\alpha \beta}{x_i}$, $\theta_i = \frac{(\alpha \beta)^\alpha w_i x_i^{-\alpha+\beta-1}}{\Gamma(\alpha) \Gamma(\beta)}$, n and $\{x_i, w_i\}_{i=1}^n$ are the Gauss-Laguerre approximation order, nodes, and weights, respectively [30, Table 25.9]. The CDF of h is given as

$$F_h(h) = \sum_{i=1}^n a_i \left[\frac{\psi_i^{\tilde{\varphi}-\alpha} h^{\tilde{\varphi}}}{\chi^{\tilde{\varphi}}} \Gamma \left(\alpha - \tilde{\varphi}, \frac{\psi_i h}{\chi} \right) + \psi_i^{-\alpha} \gamma \left(\alpha, \frac{\psi_i h}{\chi} \right) \right], \quad (5)$$

where $\gamma(\cdot, \cdot)$ is the lower incomplete Gamma function. The derivation of Eq. (5) is detailed in Appendix Appendix A. Eqs. (4) and (5) can quickly converge with $n = 30$ terms.

For the signal model, it is well-known that the fading FSO channel is slow-varying, implying that the instantaneous channel transmittance h remains constant over many symbol durations. As a result, the average number of detections and erroneous detections in a coherence time are $N_{\text{coh}} P_s(h)$ and $N_{\text{coh}} P_e(h)$, where $N_{\text{coh}} = R_b t_{\text{coh}}$ is the number of transmitted symbols in a coherence duration, R_b is the FSO channel's bit rate, and t_{coh} is the channel coherence duration. Moreover, $P_s(h)$ is the sift probability given a channel transmittance h and can be expressed as [26]

$$P_s(h) = P_{A,B}(0,0) + P_{A,B}(0,1) + P_{A,B}(1,0) + P_{A,B}(1,1), \quad (6)$$

where $P_{A,B}(a,b)$ ($a, b \in \{0,1\}$) denotes the probability that Alice transmits bit 'a' and Bob detects bit 'b', which is calculated as [26]

$$P_{A,B}(a,b) = P_A(a) Q \left[\frac{\Xi(a,b)}{\sigma_n} \right], \quad (7)$$

where $P_A(a) = 0.5$ presents the probability that Alice sends bit 'a', $\Xi(a,b) = \frac{1}{4} (1 - 2b) (2a - 1) \Re \delta P_t h + (2b - 1) d_b$,

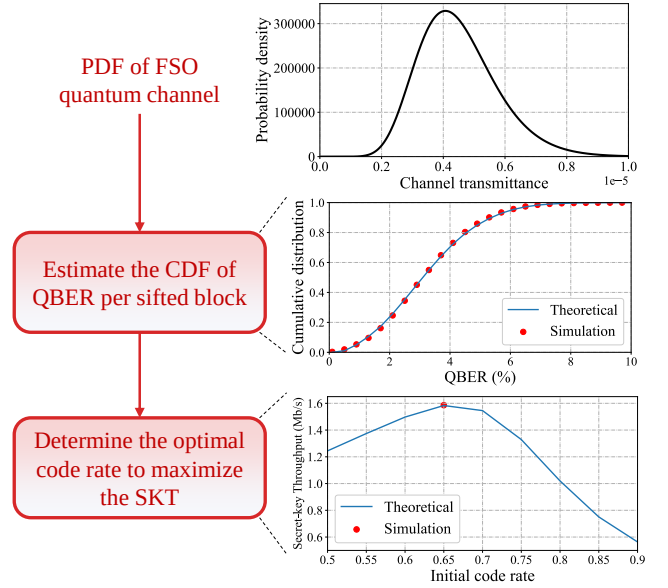


Fig. 3 Block diagram for the initial code rate selection.

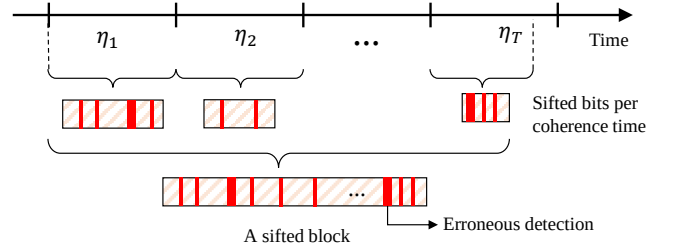


Fig. 4 Example of how a sifted block is formed over different channel coherence times.

$\Re$ is the detector's responsivity, δ is the modulation depth, P_t is the peak transmitted power, and σ_n represents standard deviation of the receiver's AWGN noise, which can be written as $\sigma_n = \sqrt{\sigma_{\text{sh}}^2 + \sigma_{\text{bg}}^2 + \sigma_{\text{th}}^2}$. Herein, σ_{sh}^2 , σ_{bg}^2 , and σ_{th}^2 are the variances of the shot noise, background noise, and thermal noise, respectively. The calculations for these factors are presented in [31]. The values of DTs are set as $d_i = (2i - 1) \left(\frac{1}{4} \Re \delta P_t h_c \frac{A_{\text{mod}} \varphi_{\text{mod}}^2}{1 + \varphi_{\text{mod}}^2} + \zeta \sqrt{\sigma_n^2} \right)$, where $i \in \{0, 1\}$, and ζ denotes the DT scale coefficient [26]. Moreover, $P_e(h)$ is the erroneous detection probability among all transmitted signals and can be given as

$$P_e(h) = P_{A,B}(0,1) + P_{A,B}(1,0). \quad (8)$$

3. Proposed KR Design to Improve the SKT

This section introduces the proposed KR design based on blind reconciliation to improve the SKT, as shown in Fig. 3. The design includes two fundamental steps. First is to construct the CDF of QBER across sifted blocks based on the PDF of the FSO quantum channel, which is presented in

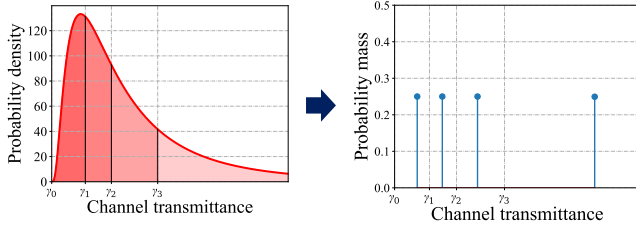


Fig. 5 Example of discretizing the channel transmittance PDF.

Sect. 3.1. Here, we assume the channel statistics are derived by observing and processing channel samples in advance [5, 32]. The second step is to theoretically determine the starting code rate that yields the highest SKT. The formulation of blind reconciliation's SKT is in Sect. 3.2, while the discussion of starting code rate selection is in Sect. 3.3. The subfigures on the right of Fig. 3 exemplify the selection of the optimal starting code rate.

3.1 Numerical Approximation of QBER Statistical Distribution

As observed, when the channel transmittance varies among different channel coherent times, the instantaneous number of detections and erroneous detections will also vary. Hence, a sifted block with a finite length can be formed over one or many channel coherence times, as depicted in Fig. 4. Without loss of generality, we assume that the collection of the sifted bits starts at the beginning of a coherence duration. As seen, the QBER of a sifted block, defined as the rate of mismatches between the same sifted blocks on both sides, is $(\sum_{t=1}^T E^{(t)})/\ell_b$. Here, T is the number of coherence durations to complete the transmission of the sifted block, $E^{(t)}$ is the number of errors arising from the t -th duration, and ℓ_b is the length of the sifted block. Note that in the case of the considered CV-QKD with DT/DD, the sifted blocks are formed over detected bits (either '0' or '1'), excluding the no detection case (X).

It can be seen that the QBER of the sifted block is a random variable. This section investigates an accurate approximation of the CDF of QBER per sifted block, formulated as

$$F_{\text{qber}}(q) = \sum_{t=1}^{\infty} F_{\text{qber},t}(q) = \sum_{t=1}^{\infty} \Pr[\text{QBER}_t \leq q], \quad (10)$$

where QBER_t is the QBER of the sifted block when it is collected in t timeslots. Consequently, QBER_t can be expressed as

$$\text{QBER}_t = \sum_{i=1}^{t-1} \frac{E^{(i)}}{\ell_b} + \frac{E^{(t)}(\ell_b - \sum_{i=1}^{t-1} D^{(i)})}{\ell_b D^{(t)}}, \quad (11)$$

where $D^{(i)}$ and $E^{(i)}$ are random variables (RV) representing

the number of detections and erroneous detections in the i -th timeslot. As a result, the probability in (10) can be rewritten as

$$F_{\text{qber},t}(q) = \Pr\left[\mathcal{E}_{t-1} + \frac{E^{(t)} D^{(t)}}{D^{(t)}} \leq q \ell_b, D^{(t)} \geq D^{(t)} > 0\right], \quad (12)$$

where $\mathcal{E}_{t-1} = \sum_{i=1}^{t-1} E^{(i)}$ and $D^{(t)} = \ell_b - \sum_{i=1}^{t-1} D^{(i)}$. It can be seen that $D^{(i)}$ and $E^{(i)}$ follows the binomial distribution. In particular, $D^{(i)} \sim \text{Bi}[N_{\text{coh}}, P_s(h^{(i)})]$, and $E^{(i)} \sim \text{Bi}[N_{\text{coh}}, P_e(h^{(i)})]$, where $h^{(i)}$ is the instantaneous channel transmittance in the i -th coherence time.

It is seen that the channel transmittance is a continuous RV. Thus, finding a closed-form expression for the statistical distribution of QBER in a timely manner is challenging, if not impossible. To derive an efficient numerical approximation, we apply the discretization method to the continuous channel transmittance, h . Specifically, we partition the domain of $f_h(h)$ into N subdomains, in which the i -th subdomain ($i \in [1, N]$) is determined in the range of $[\gamma_i, \gamma_{i+1})$. Therein, γ_j ($j \in [1, N+1]$) are called discretization thresholds, where $\gamma_1 = 0$ and $\gamma_{N+1} = \infty$. Fig. 5 illustrates the discretization of the channel PDF into four levels. The occurring probability of j -th level is given as

$$\pi_j = \int_{\gamma_j}^{\gamma_{j+1}} f_h(h) dh = F_h(\gamma_{j+1}) - F_h(\gamma_j). \quad (13)$$

Here, we select the discretization thresholds so that the probability of each level occurring is equal, i.e., $\pi_1 = \pi_2 = \dots = \pi_N$. Numerical methods are applied to determine the thresholds that satisfy this condition. Noting that in some cases, the average number of detections is larger than the sifted block length, or $N_{\text{coh}} P_s(\bar{h}_j) > \ell_b$. This means that we can obtain several sifted blocks from that channel coherence time. Consequently, the probability that a sifted block experiences the channel transmittance $\bar{h}_j$ is

$$\pi'_j = \frac{\pi_j \left[\max\left\{1, N_{\text{coh}} P_s(\bar{h}_j) / \ell_b\right\} \right]}{\sum_{i=1}^N \pi_i \left[\max\left\{1, N_{\text{coh}} P_s(\bar{h}_i) / \ell_b\right\} \right]}. \quad (14)$$

Furthermore, the channel transmittance in a coherence duration is one of N values, $\bar{h}_i$ ($1 \leq i \leq N$), which is determined by averaging over the corresponding subdomain, or

$$\bar{h}_i = \frac{\int_{\gamma_i}^{\gamma_{i+1}} h f_h(h) dh}{\pi_i} = \frac{F_2(\gamma_{i+1}) - F_2(\gamma_i)}{\pi_i}, \quad (15)$$

where $F_2(x) = \int_0^x h f_h(h) dh$, or can be calculated as

$$F_2(x) = \sum_{i=1}^n a_i \left[\psi_i^{\tilde{\varphi}-\alpha} \frac{\tilde{\varphi} x^{\tilde{\varphi}+1}}{(\tilde{\varphi}+1) \chi^{\tilde{\varphi}}} \Gamma\left(\alpha - \tilde{\varphi}, \frac{\psi_i x}{\chi}\right) \right]$$

$$F_{\text{qber},t}(q) \approx \sum_{\mathbf{k}} \sum_{m=1}^N \Pr[\mathbf{k}] \pi'_m \Pr \left[\underbrace{\mathcal{E}_{t-1|\mathbf{k}} + \frac{E_m(\ell_b - \mathcal{D}_{t-1|\mathbf{k}})}{D_m}}_{A_{t|\mathbf{k},m}} \leq q\ell_b, D_m \geq \ell_b - \mathcal{D}_{t-1|\mathbf{k}} > 0 \right]. \quad (9)$$

$$+ \psi_i^{-\alpha-1} \frac{\tilde{\varphi}\chi}{\tilde{\varphi}+1} \gamma \left(\alpha+1, \frac{\psi_i\chi}{\chi} \right). \quad (16)$$

The method to obtain (16) is the same as the one in Appendix Appendix A.

Now, we compute the approximation of $F_{\text{qber},t}$ in Eq. (12). First, let's define the channel state occurring vector $\mathbf{k} = [k_1, k_2, \dots, k_N]^T$, where k_j is the total times that the sifted block experiences the transmission coefficient $\bar{h}_j$ during the first $t-1$ timeslots. It implies that $\sum_{j=1}^N k_j = t-1$. Accordingly, the probability of the channel state occurring vector $\mathbf{k}$ is calculated as

$$\Pr[\mathbf{k}] = \binom{t-1}{k_1, k_2, \dots, k_N} \prod_{i=1}^N (\pi'_i)^{k_i}, \quad (17)$$

where $\binom{t-1}{k_1, k_2, \dots, k_N}$ is the multinomial coefficient. By conditioning on $\mathbf{k}$, we can transform (12) into (9). In this equation, D_j and E_j are RVs representing the number of detections and erroneous detections given the channel transmittance $\bar{h}_j$, i.e., $D_j \sim \text{Bi}[N_{\text{coh}}, P_s(\bar{h}_j)]$, and $E_j \sim \text{Bi}[N_{\text{coh}}, P_e(\bar{h}_j)]$. Moreover, $\mathcal{D}_{t-1|\mathbf{k}} = \sum_{j=1}^N k_j D_j$ and $\mathcal{E}_{t-1|\mathbf{k}} = \sum_{j=1}^N k_j E_j$. It is worth noting that the number of transmitted symbols in a coherence duration, N_{coh} , is considered to be large in this paper. As a result, the RV D_j is highly concentrated around its mean. We can utilize this fact to approximate the final term in (9) with a relatively small error as

$$\Pr[A_{t|\mathbf{k},m}] \approx \Pr \left[\mathcal{E}_{t-1|\mathbf{k}} + \frac{E_m \mathcal{D}'_{t-1|\mathbf{k}}}{D'_m} \leq q\ell_b, D'_m \geq \mathcal{D}'_{t-1|\mathbf{k}} > 0 \right], \quad (18)$$

where $\mathcal{D}'_{t-1|\mathbf{k}} = \ell_b - \sum_{j=1}^N k_j N_{\text{coh}} P_s(\bar{h}_j)$, $D'_m = N_{\text{coh}} P_s(\bar{h}_m)$. Additionally, E_j can be approximated as a normal RV as $E_j \sim \mathcal{N}(\mu_{E_j}, \sigma_{E_j}^2)$, where $\mu_{E_j} = N_{\text{coh}} P_e(\bar{h}_j)$, and $\sigma_{E_j}^2 = N_{\text{coh}} P_e(\bar{h}_j) [1 - P_e(\bar{h}_j)]$. Consequently, $A_{t|\mathbf{k},m}$ is the sum of t normal distributions or $A_{t|\mathbf{k},m} \sim \mathcal{N}(\mu_A, \sigma_A^2)$, where $\mu_A(t, \mathbf{k}, m) = \mu_{E_m} \frac{\mathcal{D}'_{t-1|\mathbf{k}}}{D'_m} + \sum_{i=1}^N k_i \mu_{E_i}$, $\sigma_A^2(t, \mathbf{k}, m) = \sigma_{E_m}^2 \frac{\mathcal{D}'_{t-1|\mathbf{k}}}{D'_m} + \sum_{i=1}^N k_i \sigma_{E_i}^2$. By substituting (10), (17), (9), we have

$$F_{\text{qber}}(q) \approx \sum_t \sum_{\mathbf{k}} \sum_{m=1}^N \Pr[\mathbf{k}] \pi'_m F_{A_{t|\mathbf{k},m}}(q\ell_b)$$

Algorithm 1: Pseudocode for implementing (19)

```

1 Input  $\pi' = [\pi'_1, \pi'_2, \dots, \pi'_N]^T$ ;
2 initialize  $F_{\text{qber}}(q) = 0$ ;
3 for  $t = 1 : T_{\text{max}}$  do
4   for  $\mathbf{k}$  that  $\sum_{j=1}^N k_j = t-1$  do
5      $\mathcal{D}'_{t-1|\mathbf{k}} = \ell_b - \sum_{j=1}^N k_j N_{\text{coh}} P_s(\bar{h}_j)$ ;
6     for  $m = 1 : N$  do
7       if  $D'_m \geq \mathcal{D}'_{t-1|\mathbf{k}} > 0$  then
8          $F_{\text{qber}}(q) + = \Pr[\mathbf{k}] \pi'_m F_{A_{t|\mathbf{k},m}}(q\ell_b)$ ;
9       end if
10    end for
11  end for
12 end for
13 return  $F_{\text{qber}}(q)$ ;

```

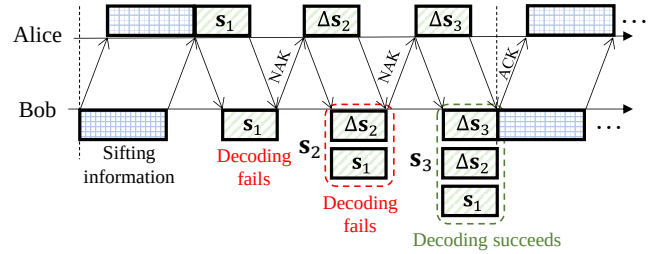


Fig. 6 Illustration of information exchanged over the public channel with the blind KR's operation.

$$\times \mathcal{I}_{(0, D'_m)}(\mathcal{D}'_{t-1|\mathbf{k}}), \quad (19)$$

where $\mathcal{I}_{\mathcal{F}}(x)$ is the indicator function, which returns one if $x \in \mathcal{F}$ and zero otherwise; $F_X(x)$ is the CDF of the normal distribution, $T_{\text{max}} = \lceil L/D_1 \rceil$ is the maximum number of considered coherence times. Algorithm 1 illustrates the pseudocode for the implementation of Eq. (19). The corresponding PDF is readily given as $f_{\text{qber}}(q) = \frac{d}{dq} F_{\text{qber}}(q)$.

3.2 Blind Reconciliation & Its SKT Formulation

3.2.1 Operations of Blind Reconciliation

The *blind reconciliation* is considered for the KR stage. In this method, both sides share a family of RC-LDPC codes that allow reusing syndrome bits of high-rate codes during the decoding of lower-rate codes [18]. In this work, we consider the blind KR whose operation is the same as the ones in [17, 22]. In particular, Alice initially creates the syndrome vector from the desired sifted block and the family's highest code rate. Bob receives the syndrome vector via the public

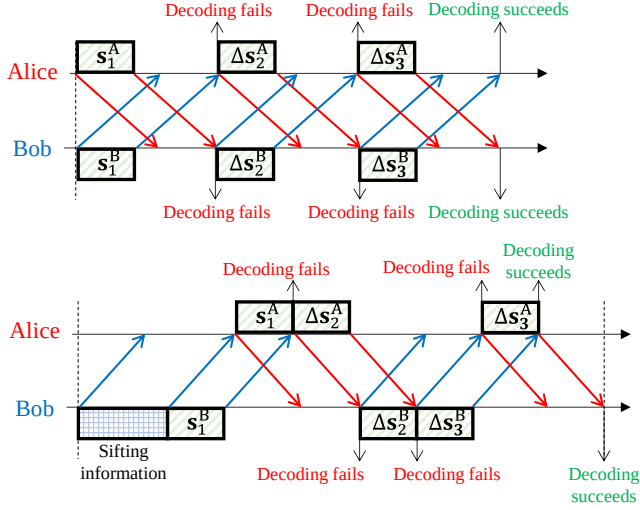


Fig. 7 Illustration of symmetric blind KR (above) and its operation in our considered system (below).

channel and corrects his block accordingly using the modified decoding algorithm. If the decoding fails, Alice creates more syndrome bits and transmits them to Bob. These bits are combined with the previous ones to form a syndrome vector for a lower-rate code. Thus, Bob can increase the chances of successful decoding on the next attempt. The procedure continues until Bob successfully decodes, or all code members in the family are used. Unsuccessfully reconciled blocks are discarded. Fig. 6 illustrates the operation of the post-processing phase, where the shifting information is transmitted first by Bob, then followed by the operation of the blind reconciliation.

The blind KR operation can also be conducted symmetrically. In particular, instead of Bob correcting his sifted blocks to Alice's one, both sides try to decode the error vector. The error vector represents the difference between both sides' sifted blocks, i.e., $e = x_A \oplus x_B$, where x_A and x_B denotes the sifted blocks of Alice and Bob, respectively. To achieve this, both users exchange their syndrome vectors and then XOR them. The newly combined vector is used to correct a zero vector into the error vector. More details of this operation can be found in [19]. However, unlike the design in [19], which discloses the true values of punctured bits on both sides, our considered methods reveal syndrome bits each round. Fig. 7 illustrates the symmetric blind KR and its operation in our considered system. As shown, the symmetric setting can potentially reduce propagation delay as no feedback of decoding results is required. However, the trade-off is increased complexity due to the need to decode at both ends.

Let N_r and R_i denote the number of code members and the i -th code rate in the RC family, respectively ($1 > R_1 > \dots > R_{N_r} > 0$). This work considers the information-theoretic approach for modeling the performance of the decoding procedure. In particular, the decoding error probability of a sifted block given the finite length of

ℓ_b can be approximated as [33]

$$\epsilon(R_i, q) \approx Q\left\{\frac{\ell_b [1 - h(q)] - \ell_b R + 0.5 \log(\ell_b)}{\sqrt{\ell_b V(q)}}\right\}, \quad (20)$$

where R_i is the considered code rate, q is the QBER of the sifted block, $Q\{\cdot\}$ denotes the Gaussian Q function, and $V(q) = q(1-q)\log^2\left(\frac{1-q}{q}\right)$ is the channel dispersion of the binary symmetric channel [33].

3.2.2 End-to-end Secret-key Throughput (SKT)

The *end-to-end SKT* of the system with the blind reconciliation can be formulated as

$$\text{SKT} = \frac{\ell_b \sum_{i=1}^{N_r} P_{\text{succ}}^{(i)} (\beta_i I_{AB} - I_{AE})}{\max[\bar{\epsilon}_Q, \bar{\epsilon}_{P,s}]}, \quad (21)$$

where $\beta_i = \frac{R_i}{I_{AB}}$ denotes the reconciliation efficiency, $P_{\text{succ}}^{(i)}$ is the fraction of sifted block reconciled by the i -th code rate, and can be expressed as

$$P_{\text{succ}}^{(i)} = \int_0^1 f_{\text{qber}}(q) [1 - \epsilon(R_{N_i}, q)] \prod_{j=1}^{i-1} \epsilon(R_{N_j}, q) dq. \quad (22)$$

Eq. (22) can be numerically obtained as

$$P_{\text{succ}}^{(i)} \approx \frac{1}{2} \sum_{k=1}^m w_k f_{\text{qber}}(\varrho_k) [1 - \epsilon(R_{N_i}, \varrho_k)] \prod_{j=1}^{i-1} \epsilon(R_{N_j}, \varrho_k), \quad (23)$$

where $\varrho_k = 0.5x_k + 0.5$, m and $\{x_k, w_k\}_{k=1}^m$ are the Gauss-Legendre approximation order, nodes, and weights, respectively [30, Table 25.4]. Moreover, I_{AE} is the upper bound on the mutual information between the sifted bits of Alice and Eve, whose value depends on Eve-Bob distance, d_E . Specifically, d_E affects the average signal-to-noise ratio at Eve, which in turn determines the amount of information she can obtain from the sifted keys. For instance, when Eve is located close to Bob (i.e., a small d_E), she is more likely to be near the beam center and therefore experiences a higher average SNR. As a result, Eve can obtain more information about the sifted bits, leading to a larger I_{AE} . Details of the computation can be found in [6, (44)].

In the denominator of Eq. (21), $\bar{\epsilon}_Q$ is the average quantum phase delay of a sifted block derived from the simulation. Moreover, $\bar{\epsilon}_{P,s}$ is the average post-processing delay of a sifted block, and $s \in \{C, S\}$ denotes the conventional and symmetric settings, respectively. As depicted in Fig. 6, it can be expressed as

$$\bar{\epsilon}_{P,s} = t_{\text{prop}} + t_{\text{trans}}^{\text{sift}} + P_f D_{N_r,s} + \sum_{i=1}^{N_r} P_{\text{succ}}^{(i)} D_{i,s}, \quad (24)$$

where $P_f = 1 - \sum_{i=1}^{N_t} P_{\text{succ}}^{(i)}$ is the probability of fail reconciliation, $t_{\text{prop}} = L/c$ denotes the propagation delay, and c is the speed of light. Moreover, $t_{\text{trans}}^{\text{sift}}$ represents the transmission delay of sifting information, and is written as

$$t_{\text{trans}}^{\text{sift}} = \frac{n_{\text{sift}}}{R_{\text{pub}}} = \frac{B_{\text{ts}} \ell_b}{R_{\text{pub}}}, \quad (25)$$

where n_{sift} is the sifting information length, B_{ts} is the number of bits used to encode the timestamp of a detection event, and R_{pub} is the public channel's data rate. In addition, $D_{i,s}$ is the average delay given that the reconciliation is finished at the i -th round and can be given as

$$D_{i,s} = \begin{cases} (2i-1)t_{\text{prop}} + t_{\text{trans}}^{\text{syn},i}, & s = C, \\ \left(2\left\lceil \frac{i}{2} \right\rceil - 1\right)t_{\text{prop}} + t_{\text{trans}}^{\text{syn},i}, & s = S, \end{cases} \quad (26)$$

where $t_{\text{trans}}^{\text{syn},i} = n_{\text{syn},i}/R_{\text{pub}}$ is the transmission delay of the i -th syndrome vector, and $n_{\text{syn},i} = \ell_b (1 - R_i)$ is the syndrome length of the code rate R_i .

3.3 Discussions on Initial Code Rate Selection

As shown in (21), to improve the SKT, the KR stage needs to (1) lower the KR efficiency and (2) reduce the post-processing delay. For blind reconciliation given an RC code family, the KR efficiency can always be optimal. The reason is that Bob tries each code member, from the highest to the lowest, until he decodes successfully. However, when the QBER is high, the blind KR may require several communication rounds to reach the proper code rate. This increases post-processing latency, resulting in a lower SKT.

To improve the SKT, one way is to start from a lower-rate code member. This decreases latency but increases KR efficiency, which eventually leads to a higher SKT. Fig. 3 shows the SKR versus different starting code rates. As observed, the optimal initial code rate is 0.65. Starting from lower-rate codes (e.g., 0.5) results in high KR efficiency, whereas higher initial code rates (e.g., 0.9) lead to higher post-processing latency. In general, we can determine the optimal initial code rate by exhaustively checking all possible inputs and selecting the one that yields the maximum SKT.

4. Numerical Results & Discussions

This section evaluates the performance of the proposed KR design under different scenario settings. Unless otherwise stated, the default values of system parameters are as follows: satellite altitude $H_s = 500$ km, zenith angle $\xi = 30^\circ$, FSO channel's bit rate $R_b = 1$ Gbps, modulation depth $\delta = 0.2$, divergence half-angle $\theta = 25 \mu\text{rad}$, $\mu_x = \mu_y = 0$, jitter angles $\theta_{\text{jt},x} = \theta_{\text{jt},y} = 2.5 \mu\text{rad}$, transmitted power $P_t = 25$ dBm, sifted block length $\ell_b = 10^4$ bits, Bob's DT scale coefficient $\zeta = 2.5$, Eve-Bob distance $d_E = 10$ m, atmospheric altitude $H_a = 20$ km, ground turbulence level $C_n^2(0) = 10^{-14} \text{ m}^{-2/3}$,

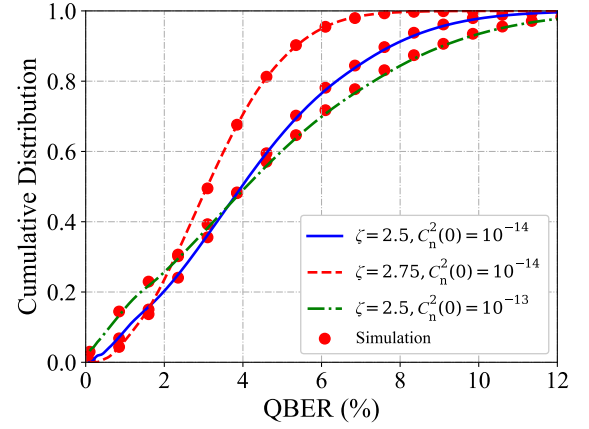


Fig. 8 Approximated CDF of sifted blocks' QBER in different scenarios.

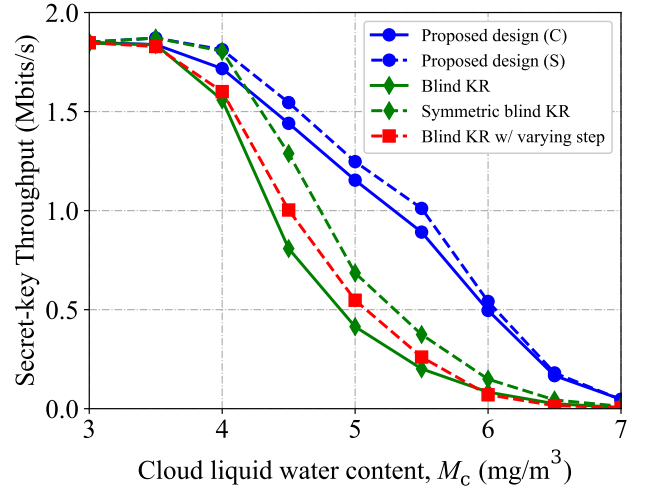


Fig. 9 SKT performance of the proposed design and blind KR strategies (C: Conventional, S: Symmetric).

coherence duration $t_{\text{coh}} = 1$ ms, vertical extent of clouds $H_c = 2$ km, CLWC $M_c = 5 \text{ mg/m}^3$, cloud droplet number concentration $N_c = 250 \text{ cm}^{-3}$, number of bits used to encode a timestamp $B_{\text{ts}} = 64$ bits, and public channel's data rate $R_{\text{pub}} = 1$ Gbps. We also consider a family of code members $\{0.9, 0.85, \dots, 0.5\}$. Other parameters can be found in [34].

First, we investigate the QBER statistical distribution with different DT scale coefficients and ground turbulence strength, as shown in Fig. 8. As seen, the simulation results closely follow the analytical line, confirming the feasibility of the numerical approximation. Moreover, when the DT scale coefficient increases from 2.5 (—) to 2.75 (---), the QBER tends to fluctuate over a narrower range. This is because a high DT scale coefficient leads to a low sift probability. As a result, a block spans over many coherence times, effectively averaging the QBER. It is also observed that a stronger turbulence strength (---) can increase the QBER fluctuation level.

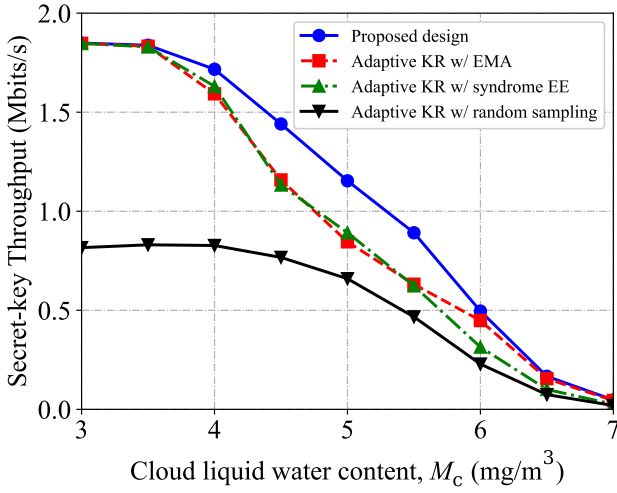


Fig. 10 SKT performance of the proposed design and rate-adaptive KR strategies.

Next, we highlight the effectiveness of the proposed designs (—●—) against existing blind KR designs in Fig. 9. Particularly, we account for conventional blind KR, symmetric blind KR, and the blind KR with varying step [20]. In addition, we consider our proposed design in both conventional and symmetric settings. The results demonstrate that our designs outperform benchmarks over the considered range. This happens because when the CLWC increases, the average QBER also increases. Consequently, blind KR designs may require several communication rounds to reach the proper code rate. By contrast, our design can adapt the initial code rate to channel conditions, resulting in significant performance improvements. Moreover, as there is no significant gap between the conventional and symmetric settings of our proposal, we adopt the conventional one for subsequent results.

To further highlight the effectiveness of our proposed design, Fig. 10 also considers the adaptive rate KR with different EE methods, i.e., exponential moving average (EMA) [24], syndrome EE [25], and random sampling [16]. The key idea of EMA is to leverage the QBER values from previous blocks to predict the current block's QBER. As for syndrome EE, Alice first transmits the syndrome bits of the highest-rate code. If decoding fails, Bob uses these syndrome bits to estimate the QBER. Both sides then select the proper code rate accordingly. In random sampling, both sides always dedicate the first round to estimate the error by exchanging a portion of their sifted bits. In this work, the portion of revealing bits is set as 15%. The thresholds for code rate selection are taken in [8]. For the syndrome EE and random sampling, we assume perfect estimation as an upper bound of the methods. As for the EMA method, we consider the RC code family so that Alice can transmit incremental syndrome bits in case of failed decoding.

As seen, the proposed designs outperform other methods over the considered range of CLWC. The EMA (—■—)

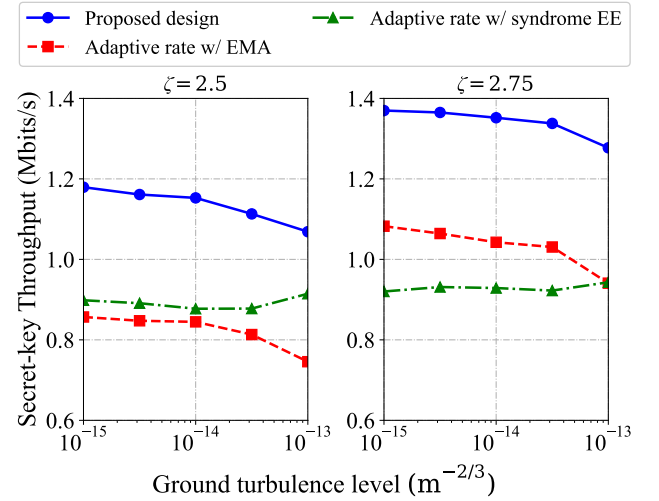


Fig. 11 SKT performance of different KR strategies versus turbulence strength for two DT scale coefficient values, ζ .

and syndrome EE (—▲—) follow closely, as the methods can sometimes complete the reconciliation within a single communication round when the QBER is low. The random sampling method (—▼—), on the other hand, always takes two communication rounds. Moreover, the number of revealed bits used to estimate the QBER is removed in the privacy amplification step. Consequently, the performance of random sampling is lower than that of other methods when $M_c < 6 \text{ mg/m}^3$.

We further demonstrate an advantage of the proposed design under the effect of small-scale fading in Fig. 11. In this figure, we plot the SKT performance of the proposed design, EMA, and the syndrome EE versus the turbulence strength in two cases of DT scale coefficients, i.e., $\zeta = 2.5$ and $\zeta = 2.75$. It can be seen that the EMA performance significantly drops when the QBER across blocks is more variable. This is because the prediction based on the QBER of previous blocks becomes less accurate in this case. For the syndrome EE, it is more resilient to the channel turbulence. However, this method is ineffective in some cases. For example, when the average QBER is high, the highest-rate code usually fails, and the reconciliation takes two RTTs. In both cases, the proposed design outperforms both benchmarks, highlighting its effectiveness in dynamic FSO channel conditions.

Next, Fig. 12 presents the effect of the user's DT scale coefficient on the SKT of the proposed design and selected benchmarks, i.e., symmetric blind KR, adaptive KR with EMA, and adaptive KR with syndrome EE. As shown, increasing ζ improves the SKT performance of the proposed design. This occurs because a higher DT scale coefficient lowers the average QBER, enabling the use of higher code rates with less information leakage. However, to some extent, excessively high values of ζ reduce the sift probability, thereby degrading SKT performance. Moreover, when $\zeta \in (1.5, 3)$, our proposed design outperforms the bench-

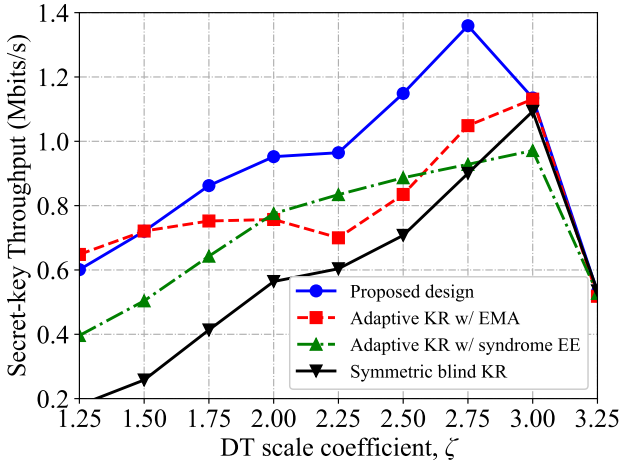


Fig. 12 SKT performance of the proposed designs and selected benchmarks with different DT scale coefficient values, ζ .

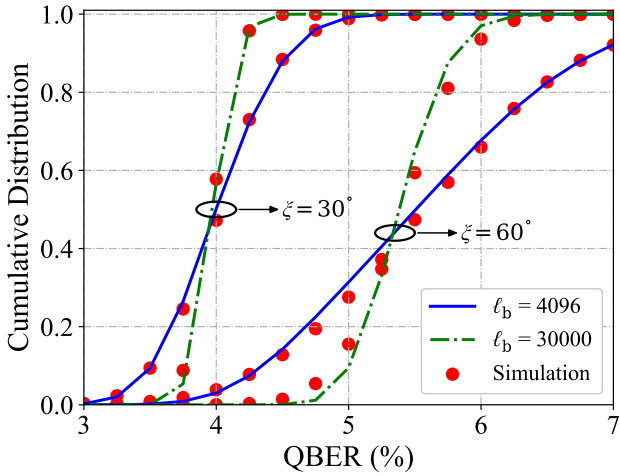


Fig. 13 QBER statistical distribution of decoy-state BB84 with different zenith angle values and block lengths.

marks. Otherwise, adaptive KR with EMA (-■-) achieves performance similar to that of our design. This occurs because when $\zeta \geq 3$ or $\zeta \leq 1.5$, the QBER values of the sifted blocks are mostly very low or very high, respectively. Hence, the EMA method yields either the highest or the lowest code rate in the family.

From this result, we can select the DT scale coefficient to maximize the SKT. For instance, ζ should be selected as 2.75 in this case. It is worth noting that joint selection of the DT scale coefficient and the initial code rate can be conducted to achieve additional SKT gain. Specifically, we additionally search over a discrete set of ζ , e.g., $\{1.25, \dots, 3.25\}$. Accordingly, we can determine a ζ and an initial code rate for further SKT enhancement.

Next, we demonstrate the feasibility of the theoretical framework for the two-decoy-state BB84 protocol. The

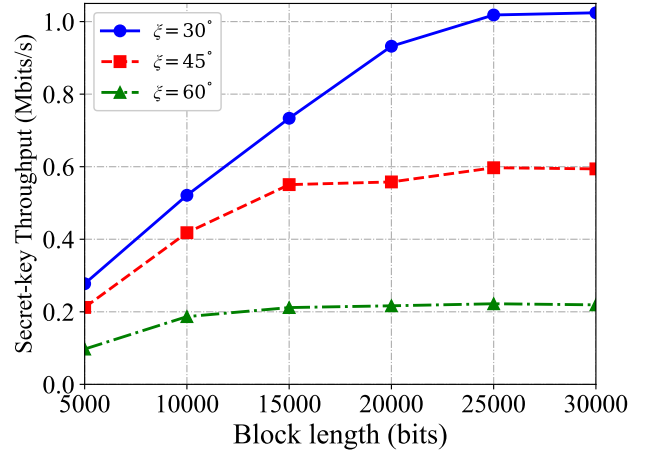


Fig. 14 SKT of decoy-state BB84 with zenith angle values and block lengths.

channel model and considered parameters are the same as in [35], with the considered ones: transmission efficiency $\tau_{\text{zen}} = 0.91$, coherence time $t_{\text{coh}} = 1$ ms, after-pulsing probability $P_{\text{AP}} = 0$, and pulse repetition rate $\mathcal{R} = 2 \times 10^9$ pulses/s. Fig. 13 presents the QBER statistical distribution at two different zenith angles, which are 30° and 60° . The results show a perfect match between the theoretical and simulation results, demonstrating the viability of our methods for the decoy-state BB84 protocol. Moreover, it is observed that as the zenith angle increases, the QBER fluctuation is greater. This is because turbulence is stronger at high-zenith-angle positions. A mitigation method is to increase the sifted block length. As a result, the QBER tends to average over multiple coherence times.

Next, Fig. 14 shows the performance of the BB84 protocol with different block lengths and zenith angle values. As observed, increasing the block length improves the SKT as more bits are processed in a reconciliation turn. However, to some extent, the SKT saturates because the time to collect a block exceeds its post-processing latency. From this result, we can determine the optimal block length to achieve the maximum SKT. For instance, the block length should be selected as 25000 bits when the zenith angle ξ is 30° .

5. Conclusion

This paper introduced a novel KR design to improve the SKT in satellite FSO/QKD systems. In particular, the design exploited the trade-off between the KR efficiency and the number of communication rounds of blind KR to determine an optimal code rate that maximizes SKT. To achieve this, we proposed an analytical framework to approximate the cumulative distribution of sifted blocks' QBER by leveraging the PDF of quantum FSO channels. We demonstrated the effectiveness of the proposed design compared to the state-of-the-art methods under various channel conditions. The feasibility of the theoretical framework was illustrated for non-coherent CV-QKD with an SC-BPSK/dual-threshold

detector and the decoy-state BB84 protocol. Finally, we investigated the optimal block length for the decoy-state BB84 protocol at different zenith angles.

Appendix A: Derivation of Eq. (5)

The CDF of channel transmittance h can be given as

$$\begin{aligned} F_h(h) &= \int_0^h f_h(x) dx \\ &= \frac{\tilde{\varphi}}{\chi^{\varphi_{\text{mod}}}} \sum_{i=1}^n a_i \psi_i^{\varphi_{\text{mod}} - \alpha} \int_0^h x^{\varphi_{\text{mod}} - 1} \Gamma\left(\alpha - \tilde{\varphi}, \frac{\psi_i x}{\chi}\right) dx. \end{aligned} \quad (\text{A} \cdot 1)$$

By applying partial integration and [36, 8.356.4], we transform the integral in (A·1) into $\frac{\chi^{\tilde{\varphi}}}{\tilde{\varphi}} \Gamma\left(\alpha - \tilde{\varphi}, \frac{\psi_i x}{\chi}\right) + \frac{1}{\tilde{\varphi}} \left(\frac{\psi_i}{\chi}\right)^{\alpha - \tilde{\varphi}} \int_0^h x^{\alpha - 1} e^{-\frac{\psi_i x}{\chi}} dx$. Using [36, 8.350.1], we can derive the CDF of h as in Eq. (5) after a few mathematical manipulations.

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